

Thursday Reading Assessment: Chapters 4-5

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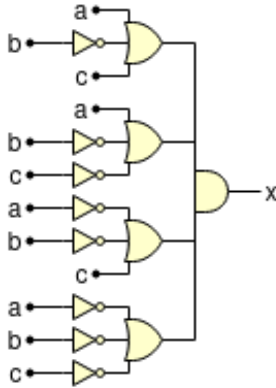


Figure 1: A logic function in need of simplification.

3. Consider the truth table below. Create the logic circuit that represents the four inputs: A, B, C, and X, the parity bit. The circuit should produce a 0 when ABC has the correct parity, and 1 when it does not, assuming even parity.

1 Logic Circuits, Boolean Algebra, and Karnaugh Maps

1. Consider the logic function in Fig. 1. (a) Based on Fig. 1, write an S-POS expression for the output. (b) From the S-POS expression, create the Karnaugh map and the truth table. (c) Generate the simplified Boolean expression from the Karnaugh map.

A	B	C	Even Parity Bit P	Parity Check X
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	0
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	0

2. Draw the simplified logic function diagram using basic gates.